



University of
Central Lancashire
UCLan Cyprus



ΣtatMod2026

Statistical Modelling with Applications

Conference Programme

18-19 September 2026
UCLan Cyprus, Larnaka, Cyprus

Welcome Message

On behalf of the Organising Committee, it is our great pleasure to welcome you to *StatMod2026: Statistical Modelling with Applications*, held at UCLan Cyprus in Larnaka on 18-19 September 2026.

Statistical modelling continues to play a fundamental role in addressing scientific and societal challenges across a broad spectrum of disciplines, including data science, artificial intelligence, engineering, finance, health sciences, environmental sciences and reliability. The aim of StatMod2026 is to provide an international forum where researchers and practitioners can present recent advances, exchange ideas and establish new collaborations.

This year's conference brings together participants from universities, research centres and industry across Europe and beyond. The programme reflects the breadth of contemporary statistical research, ranging from theoretical developments to methodological innovations and real-world applications.

The Organising Committee would like to express its sincere appreciation to all invited speakers, contributed speakers, members of the Steering Committee, Scientific Committee and reviewers for their invaluable support. We also thank our sponsors for their support and UCLan Cyprus for hosting the conference and providing an excellent environment for scientific exchange.

We hope that you will enjoy both the scientific programme and the opportunity to meet colleagues from around the world. We wish you a productive conference and a pleasant stay in Cyprus.

The Organising Committee

Sponsors

The Organising Committee gratefully acknowledges the generous support of our sponsors. Their contribution has played an important role in the successful organisation of the **StatMod2026 Conference** and is sincerely appreciated.

We thank all our sponsors for their commitment to supporting research, education, and collaboration within the mathematical and statistical sciences.

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Friday, September 18th		
9:00 - 10:00 Lobby	Conference registration	
10:00 - 10:10 AUD2	Opening Session	
	Vlad S. Barbu , Co-chair of the Σ statMod Conference Series Welcome Address Milto Hadjikyriakou , Chair of the Organising Committee of Σ statMod2026 Welcome Address	
10:10- 11:10 AUD2	Plenary Session	
	Chair: Vlad S. Barbu	
	Davy Paindaveine , Stopping on the last success with unknown odds	
11:10-11:40 Lobby	Coffee break - Group Photo	
11:40 -13:10	Statistical Methodology and Inference Chair: Bojana Milošević Room CY006	Information Measures and Statistical Divergences Chair: Vlad S. Barbu Room CY013
	Argyro Damoulaki , Bounding posterior evidence for variable screening	Aida Toma , Robust tests for moment condition models
	Dounia BOURZIK , Detecting bilinear dependence in financial panel data: a rank-based approach	Razvan-Cornel Sfetcu , Tsallis and Kaniadakis divergences between probability measures
	Marija Cuparić , Testing independence under a monotone missingness surrogate model	Elena-Corina Cipu , Measuring population complexity through entropy measures
	Ionel Popescu , Convergence of PSO and the non-Gaussian limit	Silvia Dedu , Risk models in actuarial science using entropy measures
	Angeliki Boutsis , Exponentiality tail test: a U-statistics approach	Mihaela Irma Vladescu , Transition to a green economy in Romania - population perspectives: perceptions, barriers and expectations
	Bojana Milošević , A characteristic function-based test for the multi-sample problem under random right censoring	Vlad S. Barbu , On the weighted (Φ, a) power divergence family

13:10 -14:25	Lunch	
14:25 -15:40	Stochastic Models for Risk, Finance and Decision-Making Chair: Salvatore Vergine Room CY006	Statistics for Complex Data Chair: Sophia Loizidou Room CY013
	Filippo Petroni , Stochastic reliability indicators for wind energy systems: a Markov and semi-Markov perspective	Katarzyna Szczerba , How to analyse survival data with AI-informed non-linear Cox model
	Lucianna Cananà , Optimal stopping valuation of modular advertising strategies under copula-based Markov dependence	Andreas Anastasiou , Tensor time series change-point detection in cryptocurrency network data
	Guglielmo D'Amico , Information theory and spatial risk analysis	Chiara Passamonti , A nonparametric approach for noisy compositional data
	Michele Palma , A factor-driven shared hidden Markov model for high frequency assets returns	Christophe Ley , An overview of Principal Component Analysis methods for directional data
	Salvatore Vergine , Indexed Markov chains for intraday trading with technical indicators	Sophia Loizidou , Multiple change-point detection on the circle via isolation using permutation testing
15:40 -16:10	Coffee break	
16:10 -17:25	Probability Theory, Stein's Method and Random Structures Chair: Demetres Christofides Room CY006	Applied Statistical Modelling Chair: Andreas Makrides Room CY013
	Robert Gaunt , Stein's method for the matrix normal distribution	Hassan OUKHOUYA : Classification of climate change adaptation plans (CCAP/PACC) for new Brunswick municipalities using machine learning, deep learning, and generative AI models
	Ferdinand Rapin , Sharp Stein factors for integrated Pearson distributions through kernel representation	Pavel Averin : Thestatistical engine of constraint-based causal discovery: conditional independence tests in practice
	Manuela-Simona Cojoclea , When is an observation truly an outlier? a probability-coordinate perspective	Adela Ionescu , The influence of parameters on multivariate optimization in inverse modelling problems
	Christina Savvidou , Diameter thresholds of random Cayley Graphs - I	Steliana Rodino , Data-driven modelling of complex agro-environmental systems
	Demetres Christofides , Diameter Thresholds of Random Cavley Graphs - II	

17:25 - 18:35 AUD2	Plenary Session
	Chair: Milto Hadjikyriakou
	Maria Kateri , A Divergence-Based Framework for Statistical Modeling and Reliability Analysis
19:30	Dinner

Saturday, September 19th			
10:00 -11:30	Markov Chains, Time Series & Forecasting Chair: Andreas Anastasiou Room CY006	Advances in Stochastic Models and Methods with Applications Chair: Andreas Makrides Room CY013	Reliability, Acceptance Sampling and Risk Modelling Chair: Milto Hadjikyriakou Room CY018
	Vasilios Kalaitzis , Extending Grey-Markov forecasting through second-order transition dynamics for energy mix prediction	Sandra Ferreira , A variance-stabilizing framework for higher-order cumulant estimation	Spiros Dafnis , A new family of generalized r-modified reliability systems
	Dimitris Karlis , Some spatio-temporal models for count data	D.-Jacob Economides , Perpetual american options in a jump-diffusion model under Poisson observation times	Christos Papadopoulos , A refined approach to approximating the mean number of system repairs
	Anna Nalpantidi , On joint modelling of discrete and continuous time series	Dário Ferreira , Modelling structural changes in random-effect variability for short longitudinal panels	Mohd Azri Pawan Teh , Modelling new two-sided group chain acceptance sampling plans for Gamma distribution
	Leonidas Kordalis , Fast matrix-convolution methods for semi-Markov modelling	Alex Karagrigoriou , Evaluation of seismic activity via Poisson hidden Markov models	Mohd Azri Pawan Teh , Modelling new group chain acceptance sampling plans via the minimum angle method
	Adyda Ibrahim , Evaluating multi-regime operational sensitivity in urban transit: A transition from discrete to Bayesian continuous-time Markov models	Nikolaos Galanopoulos , Extensions of Grey forecasting models for insurance loss prediction	Milto Hadjikyriakou , Stochastic ordering of dependent systems under transformation models and Archimedean copulas
	Konstantinos Fokianos , Inference and applications of multivariate integer-valued time series	Andreas Makrides , Discrete-time semi-Markov modelling of residual dynamics in Grey GM(1,1) forecasting	
11:30 -12:00	Coffee break		
12:00- 13:00 AUD2	Plenary Session		
	Chair: Konstantinos Fokianos		
	Antonio Di Crescenzo , Some results on random distortions and hazard-operator families		
13:00 -14:15	Lunch		

14:15 - 15:30	Sequential Estimation Methods and Change-Point Detection for Statistical Models Chair: Ouerdia Arkoun Room CY006	Statistical topics for Markov and Semi-Markov processes Chair: Vlad S. Barbu Room CY013
	Salim Bouzebda , Asymptotic properties of semiparametric M-estimators with multiple change points	Ana-Maria Staicu , Repeatedly spatially indexed multivariate functional data: an application to prostate cancer identification
	Alain Celisse , Early stopping for avoiding wasting time with random features	Maroua SOUIAH , Maximum-likelihood estimation for censored semi-Markov chains
	Laura Dumitrescu , Penalized sequential estimation for recurrent events with high-dimensional covariates	Ayhar Chafiâa , Kernel smoothing techniques for the transition function of semi-Markov systems
	Jean-Yves Brua , Sequential model selection for nonparametric autoregressive processes	Emmanouil Nektarios Kalligeris , Mixture kernels for non-stationary time series
	Ouerdia Arkoun , Robust adaptive efficient estimation for autoregressive Big Data models	Aguêmon Léon TEHUNGUE , Parameter estimation and state decoding in hidden drifting Markov models
15:30 -15:45	Coffee break	
15:45- 16:45	Plenary Session	
	Chair: Andreas Artemiou	
	Ana Colubi , Assessing topic changes using Latent Dirichlet Allocation	
16:45	Closing Session	
	Chair: Milto Hadjikyriakou	